

# Similarity-Based Duplicate Record Detection System

## DATA SCIENCE (24AD3107) Project Review-1



### Team Members

Allipuram Akhil – 2420030540  
P. Ruthvik Reddy – 2420030546  
Surya – 2420030440



### Under the Guidance of

Ms. Bonagani Prathusha

Assistant Professor,

Department of Computer Science and Engineering,

KLH CSE Bowrampet Campus

# Outline of Project Work

- **Focus**

- **Introduction**

- **Literature Survey**

- **Challenges**

- **Research Objectives**

- **Conclusion & Future Work**

- **References**

- **Publications Details**

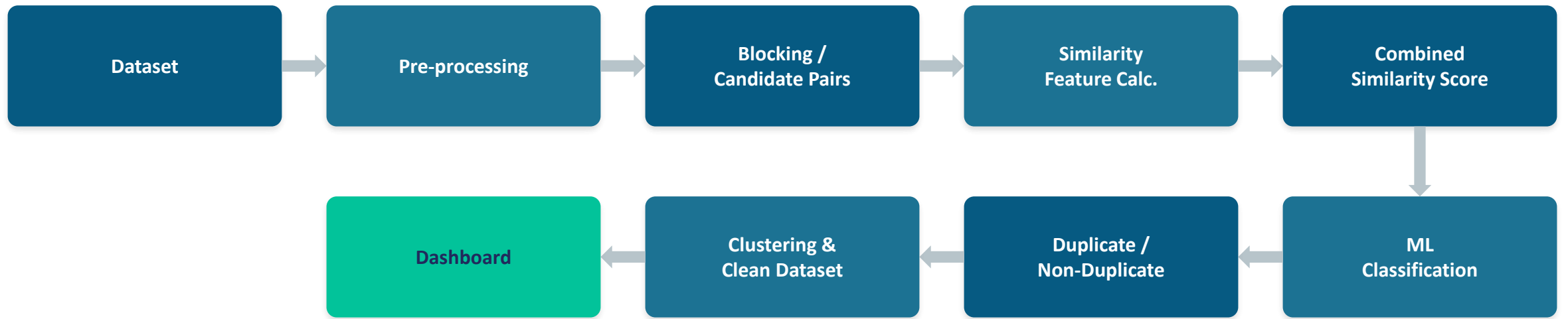
# Problem Statement

- Organizations collect customer and record data from multiple sources (web forms, system migrations, merges), which introduces exact and near-duplicate records that differ due to typos, formatting, or missing fields.
- Manual identification of duplicate records is time-consuming, error-prone, and does not scale to large datasets.
- Naively comparing every record against every other record is computationally expensive — an  $O(n^2)$  operation — and becomes impractical as data volume grows.
- There is a need for an automated system that detects both exact and near-duplicate records using similarity measures and machine learning, and produces a clean, deduplicated dataset.

# Objectives

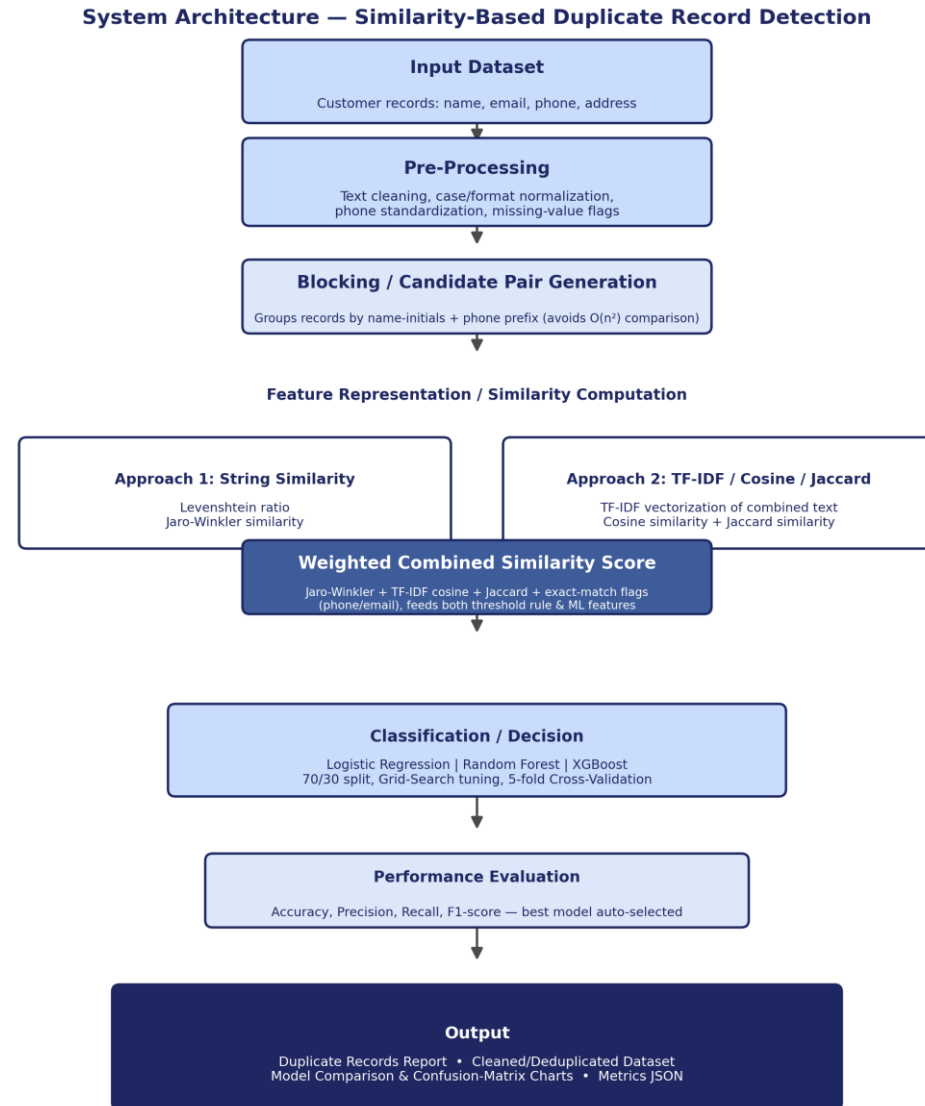
- Design a pipeline that pre-processes and standardizes raw record data — names, emails, phone numbers, and addresses.
- Implement an efficient blocking strategy that reduces the candidate-pair search space instead of comparing every record pair.
- Compute complementary similarity measures — string-based (Levenshtein ratio, Jaro-Winkler) and vector-based (TF-IDF, Cosine, Jaccard) — and combine them into a weighted similarity score.
- Train and compare supervised ML classifiers (Logistic Regression, Random Forest, XGBoost) for duplicate / non-duplicate classification, with grid-search hyperparameter tuning.
- Evaluate models using accuracy, precision, recall, and F1-score, and generate a cleaned, deduplicated output dataset.
- Provide an interactive web interface for running detection, inspecting flagged pairs, and visualizing results.

# System Workflow



*End-to-end pipeline: raw records are progressively standardized, paired, scored, classified, clustered, and surfaced through the dashboard.*

# Proposed System Architecture



# Technologies Used

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| Technology                             | Purpose                                                           |
|----------------------------------------|-------------------------------------------------------------------|
| Python                                 | Core programming and data processing                              |
| Pandas / NumPy                         | Data handling and numerical computation                           |
| Levenshtein / Jaro-Winkler libraries   | String-based similarity measures                                  |
| Scikit-learn (TF-IDF, Cosine, Jaccard) | Vector-based similarity and feature representation                |
| Scikit-learn                           | Logistic Regression, Random Forest, grid-search, cross-validation |
| XGBoost                                | Gradient-boosted classifier for duplicate/non-duplicate decision  |
| Flask                                  | Web application backend for the interactive interface             |
| HTML / CSS / JavaScript                | Front-end for the interactive dashboard                           |

# Literature Survey

| Authors                              | Title of the Paper                                                                                                        | Methodology                                                                                                          | Inferences                                                                                                                        | Limitation/Future work                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Elmagarmid, Ipeirotis & Verykios [1] | Duplicate Record Detection: A Survey, IEEE TKDE, 2007                                                                     | Comparative survey of field-matching, string-similarity, rule-based, and blocking techniques for duplicate detection | Establishes the blocking + similarity + classification taxonomy that most modern pipelines, including this one, still follow      | Survey pre-dates deep-learning approaches; does not benchmark ML classifiers on large modern datasets            |
| Ebraheem et al. [2]                  | DeepER – Deep Entity Resolution (DeepER), arXiv, 2017                                                                     | Distributed tuple representations via word embeddings, similarity scoring, and a classifier for entity resolution    | Deep representation learning reduces manual feature engineering compared to hand-crafted similarity features                      | High computational cost and needs large labelled training data; less interpretable than similarity-based scoring |
| Ravikanth, Korra & Mamidisetti [3]   | An Efficient Learning-Based Approach for Automatic Record Deduplication with Benchmark Datasets, Scientific Reports, 2024 | LSTM classifier combined with attribute-level similarity metrics, handling missing/null values across records        | Attribute-level weighting with a hybrid similarity + deep-learning classifier improves precision/recall over rule-based baselines | Higher computational overhead; accuracy can degrade on highly noisy or partially missing records                 |

# Problem Gap Identification

- Traditional duplicate detection relies on exact matching or simple rule-based string comparison, which misses near-duplicates caused by typos, reordering, or missing fields.
- Pure similarity-threshold approaches lack the discriminative power of a trained classifier and require manual threshold tuning for every new dataset.
- Many existing systems compare record pairs exhaustively, which does not scale to large, real-world datasets.
- Few systems combine blocking, multiple similarity approaches, and ML classification together with an interactive, explainable interface for verifying flagged pairs — this is the gap this project addresses.

# Innovation, Creativity, and Novelty

- Combines two complementary similarity approaches — string similarity (Levenshtein / Jaro-Winkler) and vector-based similarity (TF-IDF / Cosine / Jaccard) — into one weighted combined score, rather than relying on a single measure.
- Uses a blocking step (name-initial + phone-prefix key) so the pipeline scales to large datasets, mirroring standard practice in real-world entity resolution.
- Learns the decision boundary with ML classifiers, grid-search tuning, and 5-fold cross-validation, instead of a manually fixed similarity threshold.
- Applies union-find based cleaning to keep the most complete record from each duplicate cluster while producing a clean, deduplicated dataset.
- Ships as an interactive Flask web application with side-by-side pair comparison, live progress tracking, and downloadable reports — not just a command-line script.

# Interactive Dashboard / User Interface

## PROPOSED FLASK INTERFACE

- Dataset upload
- Duplicate detection execution
- Live progress tracking
- Duplicate pair inspection
- Side-by-side record comparison
- Similarity score display
- Results visualization
- Downloadable reports

**Duplicate Detection Dashboard (Wireframe)**

Upload Dataset (CSV)

Processing...

**Record A**

Name: ---

Email: ---

Phone: ---

**Record B**

Name: ---

Email: ---

Phone: ---

Similarity Score: 0.XX (proposed layout — illustrative mockup, not an actual screenshot or real result)

# Expected Outcome

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- Automated detection of exact and near-duplicate records
- Reduced manual effort in identifying duplicates
- Efficient candidate generation through blocking
- Similarity-based feature extraction combining string and vector-based measures
- ML-based classification for the duplicate / non-duplicate decision
- Duplicate clustering and a cleaned, deduplicated dataset
- Reports and interactive visualization through the web interface

*All items above are expected outcomes of the proposed system; experimental results are not yet available at this review stage.*

# Future Scope

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- More advanced entity-resolution models beyond the current classical ML classifiers
- Scaling to larger, real-world datasets
- More robust and adaptive blocking techniques
- Deep learning / embedding-based similarity representations
- Improved explainability of duplicate decisions
- Cloud deployment of the web application
- Real-time duplicate detection for streaming data
- Integration with enterprise databases and CRM systems

# Feasibility and Project Plan

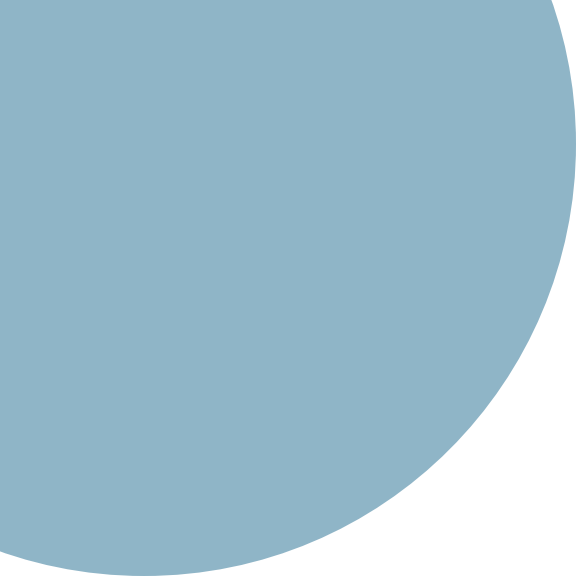
**The project is executed across six phases over the semester, from requirement study through final testing and documentation.**

- Phase 1 (Weeks 1–2): Requirement study, literature survey, and dataset design.
- Phase 2 (Weeks 3–4): Pre-processing and blocking implementation.
- Phase 3 (Weeks 5–6): Feature engineering — string similarity, TF-IDF/Cosine/Jaccard, and the combined score.
- Phase 4 (Weeks 7–8): ML model training, hyperparameter tuning, and evaluation.
- Phase 5 (Weeks 9–10): Web interface development and pipeline integration.
- Phase 6 (Weeks 11–12): Testing, documentation, and final report / viva preparation.

# References

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- [1] A. K. Elmagarmid, P. G. Ipeirotis, and V. S. Verykios, "Duplicate Record Detection: A Survey," IEEE Transactions on Knowledge and Data Engineering, 2007.
- [2] M. Ebraheem, S. Thirumuruganathan, S. Joty, M. Ouzzani, and N. Tang, "DeepER – Deep Entity Resolution," arXiv, 2017.
- [3] Ravikanth, Korra, and Mamidisetti, "An Efficient Learning-Based Approach for Automatic Record Deduplication with Benchmark Datasets," Scientific Reports, 2024.



# Thank You!

Similarity-Based Duplicate Record Detection System | Questions & Discussion